

Chapter 1: Mathematical Foundations of Unconstrained Optimisation

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Dr Dante Kalise
Department of Mathematics
Imperial College London
dkaliseb@ic.ac.uk

This week we will study unconstrained optimization problems in $\mathbb{R}^n$. We first define global and local minima (and maxima) of a function, and then we give a first characterization of stationary points of a function. Since the characterization of minima and maxima of a function is related to its Hessian, we need to study some relevant matrix properties, such as positive definiteness and diagonal dominance. These properties will be useful for stating second order and global optimality conditions. We conclude this week by studying a very important class of problems where the cost function is a quadratic form.

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Global Minimum and Maximum

Definition (Global Minimum and Maximum). Let $f : S \rightarrow \mathbb{R}$ be defined on a set $S \subseteq \mathbb{R}^n$. Then:

1. $\mathbf{x}^* \in S$ is a global minimum point of f over S if $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for any $\mathbf{x} \in S$.
2. $\mathbf{x}^* \in S$ is a strict global minimum point of f over S if $f(\mathbf{x}) > f(\mathbf{x}^*)$ for any $\mathbf{x}^* \neq \mathbf{x} \in S$.
3. $\mathbf{x}^* \in S$ is a global maximum point of f over S if $f(\mathbf{x}) \leq f(\mathbf{x}^*)$ for any $\mathbf{x} \in S$.
4. $\mathbf{x}^* \in S$ is a strict global maximum point of f over S if $f(\mathbf{x}) < f(\mathbf{x}^*)$ for any $\mathbf{x}^* \neq \mathbf{x} \in S$.



Some definitions:

- We denote by **global optimum** the global minimum or maximum.
- The maximal value of f over S :

$$\sup\{f(\mathbf{x}) : \mathbf{x} \in S\}.$$

- The minimal value of f over S :

$$\inf\{f(\mathbf{x}) : \mathbf{x} \in S\}.$$

Note that the minimal and maximal values are always unique.

Example: Find the global minimum and maximum points of $f(x_1, x_2) = x_1 + x_2$ over the unit ball in $\mathbb{R}^2$, $S = B[0, 1] = \{(x_1, x_2)^\top : x_1^2 + x_2^2 \leq 1\}$.

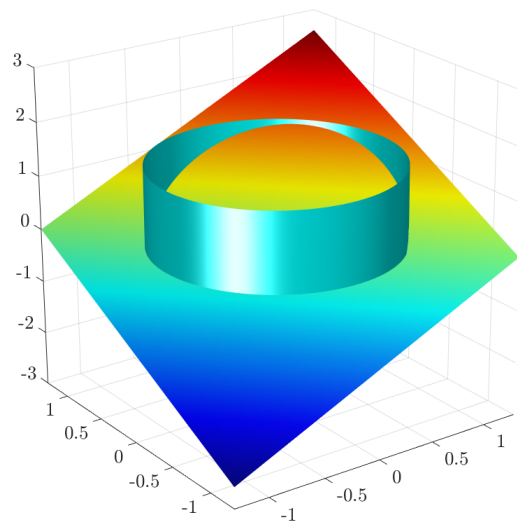


Figure 1: The function $f(x_1, x_2) = x_1 + x_2$ constrained over the unit ball.

Another example: What about the global maximizers and minimizers in $\mathbb{R}^2$ for

$$f(x_1, x_2) = \frac{x_1 + x_2}{x_1^2 + x_2^2 + 1} ?$$



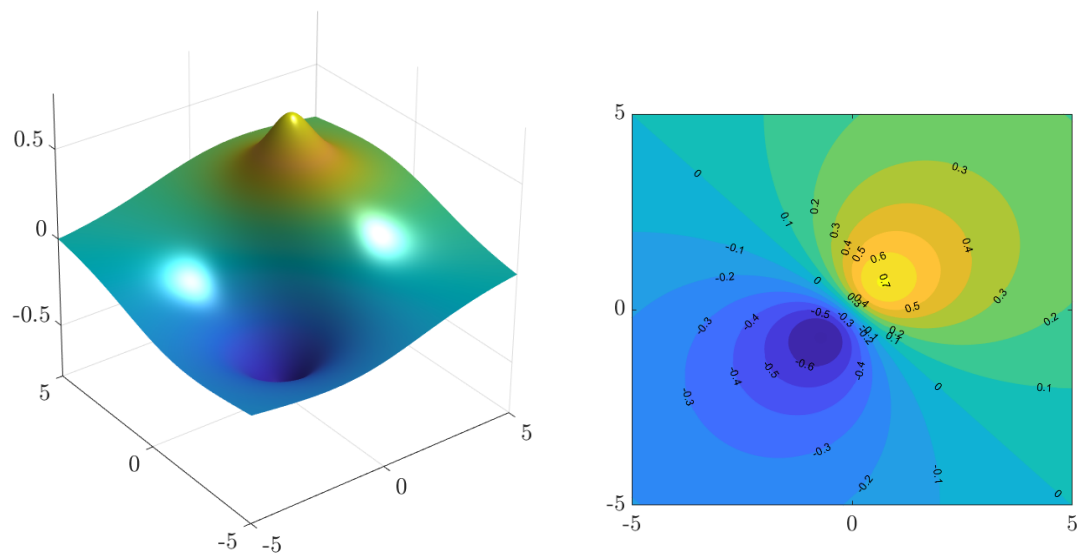


Figure 2: The function $f(x_1, x_2) = \frac{x_1 + x_2}{x_1^2 + x_2^2 + 1}$ (left) and its contour plot (right).

Local Minima and Maxima

Definition (Local Minimum and Maximum). Let $f : S \rightarrow \mathbb{R}$ be defined on a set $S \subseteq \mathbb{R}^n$. Then:

1. $\mathbf{x}^* \in S$ is a local minimum of f over S if there exists $r > 0$ for which $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for any $\mathbf{x} \in S \cap B(\mathbf{x}^*, r)$.
2. $\mathbf{x}^* \in S$ is a strict local minimum of f over S if there exists $r > 0$ for which $f(\mathbf{x}^*) < f(\mathbf{x})$ for any $\mathbf{x} \neq \mathbf{x}^*$ in $S \cap B(\mathbf{x}^*, r)$.
3. $\mathbf{x}^* \in S$ is a local maximum of f over S if there exists $r > 0$ for which $f(\mathbf{x}^*) \geq f(\mathbf{x})$ for any $\mathbf{x} \in S \cap B(\mathbf{x}^*, r)$.
4. $\mathbf{x}^* \in S$ is a strict local maximum of f over S if there exists $r > 0$ for which $f(\mathbf{x}^*) > f(\mathbf{x})$ for any $\mathbf{x} \neq \mathbf{x}^*$ in $S \cap B(\mathbf{x}^*, r)$.

Of course, a global minimum (maximum) point is also a local minimum (maximum) point.



Example: Identify the different (strict) local minima and maxima of the function

$$f(x) = \begin{cases} (x-1)^2 + 2, & -1 \leq x \leq 1 \\ 2, & 1 \leq x \leq 2 \\ -(x-2)^2 + 2, & 2 \leq x \leq 2.5 \\ (x-3)^2 + 1.5, & 2.5 \leq x \leq 4 \\ -(x-5)^2 + 3.5, & 4 \leq x \leq 6 \\ -2x + 14.5, & 6 \leq x \leq 6.5 \\ 2x - 11.5, & 6.5 \leq x \leq 8 \end{cases} \quad (1)$$

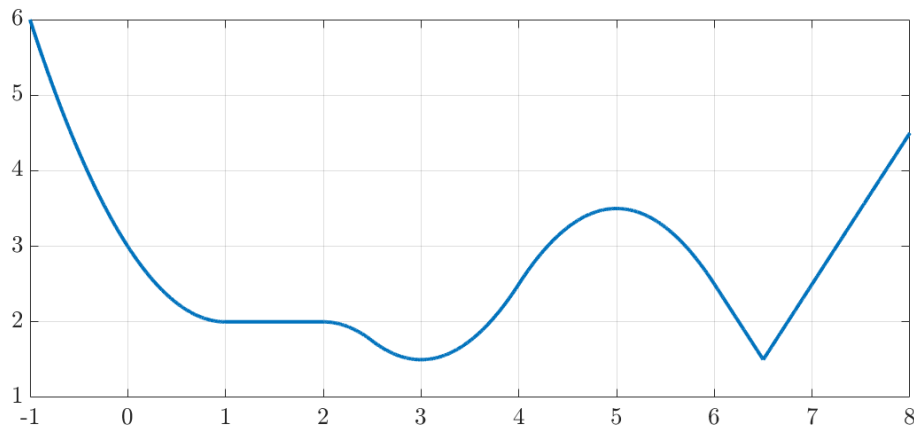


Figure 3: The function described in (1) has several minima and maxima.

Theorem (Fermat's Theorem: First Order Optimality Conditions). *Let $f : U \rightarrow \mathbb{R}$ be a function defined on a set $U \subseteq \mathbb{R}^n$. Suppose that $\mathbf{x}^* \in \text{int}(U)$ is a local optimum point and that all the partial derivatives of f exist at $\mathbf{x}^*$. Then $\nabla f(\mathbf{x}^*) = 0$.*

Proof. Let $i \in \{1, 2, \dots, n\}$ and consider the 1-D function $g(t) = f(\mathbf{x}^* + t\mathbf{e}_i)$. Then, $\mathbf{x}^*$ is a local optimum point of f implies that $t = 0$ is a local optimum of g , and hence $g'(0) = 0$. Thus, $\frac{\partial f}{\partial x_i}(\mathbf{x}^*) = g'(0) = 0$.

Definition (Stationary Points). *Let $f : U \rightarrow \mathbb{R}$ be a function defined on a set $U \subseteq \mathbb{R}^n$. Suppose that $\mathbf{x}^* \in \text{int}(U)$ and that all the partial derivatives of f are defined at $\mathbf{x}^*$. Then $\mathbf{x}^*$ is called a stationary point of f if $\nabla f(\mathbf{x}^*) = 0$.*

Example:

$$\min \left\{ f(x_1, x_2) = \frac{x_1 + x_2}{x_1^2 + x_2^2 + 1} : x_1, x_2 \in \mathbb{R} \right\}$$

$$\nabla f(x_1, x_2) = \frac{1}{(x_1^2 + x_2^2 + 1)^2} \begin{pmatrix} (x_1^2 + x_2^2 + 1) - 2(x_1 + x_2)x_1 \\ (x_1^2 + x_2^2 + 1) - 2(x_1 + x_2)x_2 \end{pmatrix}$$





Figure 4: Pierre de Fermat (1607-1665), French mathematician (and lawyer!) who made significant contributions to differential calculus and number theory.

Stationary points are those satisfying:

$$\begin{aligned} -x_1^2 - 2x_1x_2 + x_2^2 &= -1 \\ x_1^2 - 2x_1x_2 - x_2^2 &= -1. \end{aligned}$$

Hence, the stationary points are $(1/\sqrt{2}, 1/\sqrt{2})$ and $(-1/\sqrt{2}, -1/\sqrt{2})$, with $(1/\sqrt{2}, 1/\sqrt{2})$ a global maximum, and $(-1/\sqrt{2}, -1/\sqrt{2})$ a global minimum.

Second Order Optimality Conditions

Theorem (Necessary Second Order Optimality Conditions). *Let $f : U \rightarrow \mathbb{R}$ be a function defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is twice continuously differentiable over U and that $\mathbf{x}^*$ is a stationary point. Then*

1. *if $\mathbf{x}^*$ is a local minimum point, then $\nabla^2 f(\mathbf{x}^*) \geq 0$.*
2. *if $\mathbf{x}^*$ is a local maximum point, then $\nabla^2 f(\mathbf{x}^*) \leq 0$.*

Proof. We prove 1. There exists a ball $B(\mathbf{x}^*, r) \subseteq U$ for which $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for all $\mathbf{x} \in B(\mathbf{x}^*, r)$. Next, let $\mathbf{d} \in \mathbb{R}^n$ be a nonzero vector. For any $0 < \alpha < \frac{r}{\|\mathbf{d}\|}$, we have $\mathbf{x}_\alpha^* \equiv \mathbf{x}^* + \alpha\mathbf{d} \in B(\mathbf{x}^*, r)$ and for any such α , $f(\mathbf{x}_\alpha^*) \geq f(\mathbf{x}^*)$.

On the other hand, there exists a vector $\mathbf{z}_\alpha \in [\mathbf{x}^*, \mathbf{x}_\alpha^*]$ such that

$$f(\mathbf{x}_\alpha^*) - f(\mathbf{x}^*) = \frac{\alpha^2}{2} \mathbf{d}^\top \nabla^2 f(\mathbf{z}_\alpha) \mathbf{d}.$$

This implies that for any $\alpha \in (0, \frac{r}{\|\mathbf{d}\|})$ the inequality $\mathbf{d}^\top \nabla^2 f(\mathbf{z}_\alpha) \mathbf{d} \geq 0$ holds. Since $\mathbf{z}_\alpha \rightarrow \mathbf{x}^*$ as $\alpha \rightarrow 0^+$, we obtain that $\mathbf{d}^\top \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$, which leads to $\nabla^2 f(\mathbf{x}^*) \geq 0$. The proof of 2. follows analogously. $\square$



Theorem (Sufficient Second Order Optimality Conditions). *Let $f : U \rightarrow \mathbb{R}$ be a function defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is twice continuously differentiable over U and that $\mathbf{x}^*$ is a stationary point. Then*

1. *if $\nabla^2 f(\mathbf{x}^*) > 0$, then $\mathbf{x}^*$ is a strict local minimum point of f over U .*
2. *if $\nabla^2 f(\mathbf{x}^*) < 0$, then $\mathbf{x}^*$ is a strict local maximum point of f over U .*

Proof. We prove 1. There exists a ball $B(\mathbf{x}^*, r) \subseteq U$ for which $\nabla^2 f(\mathbf{x}) > 0$ for any $\mathbf{x} \in B(\mathbf{x}^*, r)$. Then, by the Linear Approximation Theorem (week 1), there exists a vector $\mathbf{z}_x \in [\mathbf{x}^*, \mathbf{x}]$ (and hence $\mathbf{z}_x \in B(\mathbf{x}^*, r)$) for which

$$f(\mathbf{x}) - f(\mathbf{x}^*) = \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^\top \nabla^2 f(\mathbf{z}_x) (\mathbf{x} - \mathbf{x}^*) .$$

From this, it follows that $\nabla^2 f(\mathbf{z}_x) > 0$, implying that for any $\mathbf{x} \in B(\mathbf{x}^*, r)$ such that $\mathbf{x} \neq \mathbf{x}^*$, the inequality $f(\mathbf{x}) > f(\mathbf{x}^*)$ holds. Therefore $\mathbf{x}^*$ is a strict local minimum point of f over U . $\square$

Saddle Points

Definition (Saddle Point). *Let $f : U \rightarrow \mathbb{R}$ be a continuously differentiable function defined on an open set $U \subseteq \mathbb{R}^n$. A stationary point $\mathbf{x}^* \in U$ is called a saddle point of f over U if it is neither a local minimum point nor a local maximum point of f over U .*

Theorem (Sufficient Condition for Saddle Points). *Let $f : U \rightarrow \mathbb{R}$ be a function defined on an open set $U \subseteq \mathbb{R}^n$. Suppose that f is twice continuously differentiable over U and that $\mathbf{x}^*$ is a stationary point. If $\nabla^2 f(\mathbf{x}^*)$ is an indefinite matrix, then $\mathbf{x}^*$ is a saddle point of f over U .*

Proof. The Hessian $\nabla^2 f(\mathbf{x}^*)$ has at least one positive eigenvalue $\lambda > 0$, corresponding to a normalized eigenvector denoted by $\mathbf{v}$. There exists a radius $r > 0$ such that $\mathbf{x}^* + \alpha \mathbf{v} \in U$ for any $\alpha \in (0, r)$. By the Quadratic Approximation Theorem (week 1), there exists a function $g : \mathbb{R}_{++} \rightarrow \mathbb{R}$ satisfying

$$\frac{g(t)}{t} \rightarrow 0 \text{ as } t \rightarrow 0 ,$$

such that for any $\alpha \in (0, r)$

$$f(\mathbf{x}^* + \alpha \mathbf{v}) = f(\mathbf{x}^*) + \frac{\lambda \alpha^2}{2} \|\mathbf{v}\|^2 + g(\|\mathbf{v}\|^2 \alpha^2) .$$

Recalling that $\mathbf{v}$ is normalized, we write

$$f(\mathbf{x}^* + \alpha \mathbf{v}) = f(\mathbf{x}^*) + \frac{\lambda \alpha^2}{2} + g(\alpha^2)$$



By the properties of g , it follows that there exists an $\varepsilon_1 \in (0, r)$ such that $g(\alpha^2) > -\frac{\lambda}{2}\alpha^2$ for all $\alpha \in (0, \varepsilon_1)$ and hence $f(\mathbf{x}^* + \alpha \mathbf{v}) > f(\mathbf{x}^*)$ for all $\alpha \in (0, \varepsilon_1)$. This shows that $\mathbf{x}^*$ cannot be a local maximum point of f over U . A similar argument-exploiting an eigenvector of $\nabla^2 f(\mathbf{x}^*)$ corresponding to a negative eigenvalue-shows that $\mathbf{x}^*$ cannot be a local minimum point of f over U , establishing the desired result that $\mathbf{x}^*$ is a saddle point. $\square$

Attainment of Minimal/Maximal Points

Theorem (Weierstrass' Theorem). *Let f be a continuous function defined over a nonempty compact set $C \subseteq \mathbb{R}^n$. Then there exists a global minimum point of f over C and a global maximum point of f over C .*



Figure 5: Karl Weierstrass (1815-1897), German mathematician considered one of the founding figures of modern analysis. He formalized the definition of continuity of a function! If you want to pursue a research career in analysis, you might want to work/visit the [Weierstrass Institute](#) in Berlin.

When the underlying set is not compact, Weierstrass theorem does not guarantee the attainment of the solution, but certain properties of the function f can imply attainment of the solution.

Definition (Coerciveness). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function over $\mathbb{R}^n$. f is called coercive if*

$$\lim_{\|\mathbf{x}\| \rightarrow \infty} f(\mathbf{x}) = \infty$$

Theorem (Attainment of Global Optima Points for Coercive Functions). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous and coercive function and let $S \subseteq \mathbb{R}^n$ be a nonempty closed set. Then f attains a global minimum point on S .*



Proof. Let $\mathbf{x}_0 \in S$ be an arbitrary point in S . Since the function is coercive, it follows that there exists an $M > 0$ such that

$$f(\mathbf{x}) > f(\mathbf{x}_0) \text{ for any } \mathbf{x} \text{ such that } \|\mathbf{x}\| > M.$$

Since any global minimizer $\mathbf{x}^*$ of f over S satisfies $f(\mathbf{x}^*) \leq f(\mathbf{x}_0)$, it follows that the set of global minimizers of f over S is the same as the set of global minimizers of f over $S \cap B[0, M]$. The set $S \cap B[0, M]$ is compact and nonempty, and thus by the Weierstrass theorem, there exists a global minimizer of f over $S \cap B[0, M]$ and hence also over S . $\square$

Example. Classify the stationary points of the function $f(x_1, x_2) = -2x_1^2 + x_1x_2^2 + 4x_1^4$. The gradient is given by

$$\nabla f(x) = \begin{pmatrix} -4x_1 + x_2^2 + 16x_1^3 \\ 2x_1x_2 \end{pmatrix},$$

and the stationary points are the solutions to

$$\begin{aligned} -4x_1 + x_2^2 + 16x_1^3 &= 0 \\ 2x_1x_2 &= 0 \end{aligned}.$$

The stationary points are $(0, 0)$, $(0.5, 0)$, and $(-0.5, 0)$. We compute the Hessian

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} -4 + 48x_1^2 & 2x_2 \\ 2x_2 & 2x_1 \end{pmatrix},$$

and evaluating at the stationary points leads to

$$\nabla^2 f(0.5, 0) = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}, \quad \nabla^2 f(-0.5, 0) = \begin{pmatrix} 8 & 0 \\ 0 & -1 \end{pmatrix}, \quad \nabla^2 f(0, 0) = \begin{pmatrix} -4 & 0 \\ 0 & 0 \end{pmatrix}.$$

Classify each stationary point.



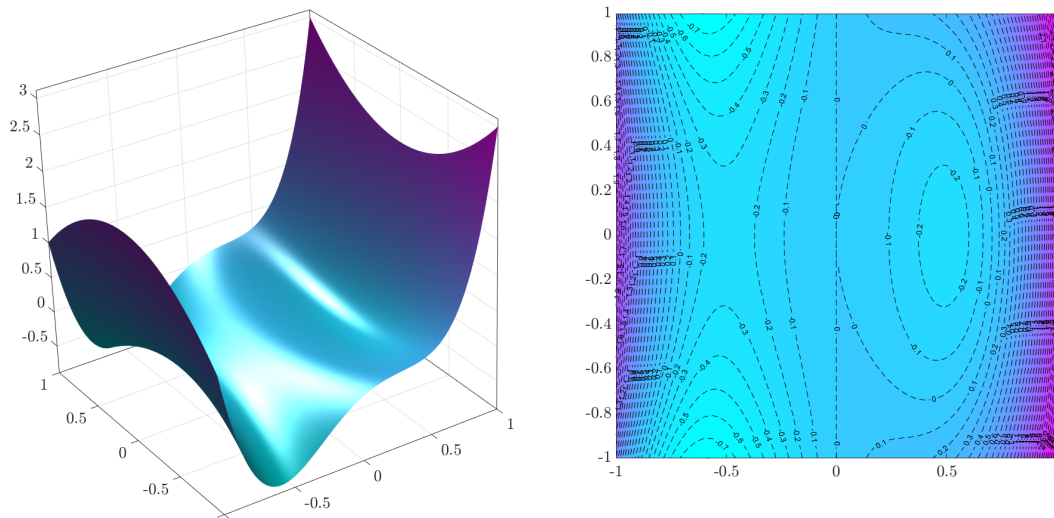


Figure 6: The function $f(x_1, x_2) = -2x_1^2 + x_1x_2^2 + 4x_1^4$ and its contour plot.

Global Optimality Conditions

Theorem (Global Optimality Condition). *Let f be a twice continuously differentiable function defined over $\mathbb{R}^n$. Suppose that $\nabla^2 f(\mathbf{x}) \geq 0$ for any $\mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{x}^* \in \mathbb{R}^n$ be a stationary point of f . Then $\mathbf{x}^*$ is a global minimum point of f .*

Proof. By the Linear Approximation Theorem, it follows that for any $\mathbf{x} \in \mathbb{R}^n$, there exists a vector $\mathbf{z}_x \in [\mathbf{x}^*, \mathbf{x}]$ for which

$$f(\mathbf{x}) - f(\mathbf{x}^*) = \frac{1}{2} (\mathbf{x} - \mathbf{x}^*)^\top \nabla^2 f(\mathbf{z}_x) (\mathbf{x} - \mathbf{x}^*) .$$

Since $\nabla^2 f(\mathbf{z}_x) \geq 0$, we have that $f(\mathbf{x}) \geq f(\mathbf{x}^*)$, establishing the fact that $\mathbf{x}^*$ is a global minimum point of f . $\square$

Example. The function $f(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3 + (x_1^2 + x_2^2 + x_3^2)^2$ has a gradient

$$\nabla f(\mathbf{x}) = \begin{pmatrix} 2x_1 + x_2 + x_3 + 4x_1(x_1^2 + x_2^2 + x_3^2) \\ 2x_2 + x_1 + x_3 + 4x_2(x_1^2 + x_2^2 + x_3^2) \\ 2x_3 + x_1 + x_2 + 4x_3(x_1^2 + x_2^2 + x_3^2) \end{pmatrix} ,$$

and the Hessian

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 2 + 4(x_1^2 + x_2^2 + x_3^2) + 8x_1^2 & 1 + 8x_1x_2 & 1 + 8x_1x_3 \\ 1 + 8x_1x_2 & 2 + 4(x_1^2 + x_2^2 + x_3^2) + 8x_2^2 & 1 + 8x_2x_3 \\ 1 + 8x_1x_3 & 1 + 8x_2x_3 & 2 + 4(x_1^2 + x_2^2 + x_3^2) + 8x_3^2 \end{pmatrix} .$$

We observe that $\mathbf{x} = 0$ is a stationary point. Check that $\nabla^2 f(\mathbf{x}) \geq 0$ for all $\mathbf{x}$. It follows that $\mathbf{x} = 0$ is the global minimum point.



Quadratic Functions

A **quadratic function** over $\mathbb{R}^n$ is a function of the form

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + \mathbf{c}$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, $\mathbf{b} \in \mathbb{R}^n$ and $\mathbf{c} \in \mathbb{R}$. From here onwards, we adopt the convention of assuming $\mathbf{A}$ to be symmetric.¹

Check(!) that

$$\begin{aligned}\nabla f(\mathbf{x}) &= 2\mathbf{A}\mathbf{x} + 2\mathbf{b} \\ \nabla^2 f(\mathbf{x}) &= 2\mathbf{A}\end{aligned}$$

Proposition. Let $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + \mathbf{c}$, with $\mathbf{A} \in \mathbb{R}^{n \times n}$ symmetric, $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{c} \in \mathbb{R}$. Then

1. $\mathbf{x}$ is a stationary point of f iff $\mathbf{A}\mathbf{x} = -\mathbf{b}$.
2. if $\mathbf{A} \geq 0$, then $\mathbf{x}$ is a global minimum point of f iff $\mathbf{A}\mathbf{x} = -\mathbf{b}$.
3. if $\mathbf{A} > 0$, then $\mathbf{x} = -\mathbf{A}^{-1}\mathbf{b}$ is a strict global minimum point of f .

Proof. 1. The proof follows immediately from the formula of the gradient of f .
2. since $\nabla^2 f(\mathbf{x}) = 2\mathbf{A} \geq 0$, it follows from global optimality conditions that the global minimum points are exactly the stationary points, which combined with part 1. implies the result.
3. When $\mathbf{A} > 0$, the vector $\mathbf{x} = -\mathbf{A}^{-1}\mathbf{b}$ is the unique solution to $\mathbf{A}\mathbf{x} = -\mathbf{b}$, and hence by parts 1. and 2., it is the unique global minimum point of f .

□

Two Important Theorems on Quadratic Functions

Lemma (Coerciveness of Quadratic Functions). Let $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + \mathbf{c}$ where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, $\mathbf{b} \in \mathbb{R}^n$ and $\mathbf{c} \in \mathbb{R}$. Then f is coercive if and only if $\mathbf{A} > 0$.

Theorem (Characterization of the Nonnegativity of Quadratic Functions). Consider the quadratic form $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + \mathbf{c}$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, $\mathbf{b} \in \mathbb{R}^n$ and $\mathbf{c} \in \mathbb{R}$. Then the following two claims are equivalent:

1. $f(\mathbf{x}) \equiv \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + \mathbf{c} \geq 0$, for all $\mathbf{x} \in \mathbb{R}^n$.
2. The augmented matrix $\begin{pmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^\top & \mathbf{c} \end{pmatrix} \geq 0$.

¹ Otherwise, work with $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{W} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + \mathbf{c}$, where $\mathbf{W} = \frac{\mathbf{A} + \mathbf{A}^\top}{2}$ (it's the same!).



Proof. Suppose that 2. holds. Then, in particular for any $\mathbf{x} \in \mathbb{R}^n$ the inequality

$$\begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix}^\top \begin{pmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^\top & c \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix} \geq 0$$

holds, which is the same as the inequality $\mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c \geq 0$, proving the validity of 1. Now, assume that 1. holds. We begin by showing that $\mathbf{A} \geq 0$. Suppose in contradiction that $\mathbf{A}$ is not positive semidefinite. Then there exists an eigenvector $\mathbf{v}$ corresponding to a negative eigenvalue $\lambda < 0$ of $\mathbf{A}$:

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v}.$$

Thus, for any $\alpha \in \mathbb{R}$

$$f(\alpha \mathbf{v}) = \lambda \|\mathbf{v}\|^2 \alpha^2 + 2(\mathbf{b}^\top \mathbf{v}) \alpha + c \rightarrow -\infty$$

as $\alpha \rightarrow -\infty$, contradicting the nonnegativity of f .

Now our objective is to prove 2.; that is, we want to show that for any $\mathbf{y} \in \mathbb{R}^n$ and $t \in \mathbb{R}$

$$\begin{pmatrix} \mathbf{y} \\ t \end{pmatrix}^\top \begin{pmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^\top & c \end{pmatrix} \begin{pmatrix} \mathbf{y} \\ t \end{pmatrix} \geq 0$$

which is equivalent to

$$\mathbf{y}^\top \mathbf{A} \mathbf{y} + 2t \mathbf{b}^\top \mathbf{y} + ct^2 \geq 0 \quad (2)$$

To show the validity of eq. (2) above for any $\mathbf{y} \in \mathbb{R}^n$ and $t \in \mathbb{R}$, we consider two cases. If $t = 0$, then (2) reads as $\mathbf{y}^\top \mathbf{A} \mathbf{y} \geq 0$, which is a valid inequality since we have shown that $\mathbf{A} \geq 0$. The second case is when $t \neq 0$. To show that (2) holds in this case, note that (2) is the same as the inequality

$$t^2 f\left(\frac{\mathbf{y}}{t}\right) = t^2 \left[\left(\frac{\mathbf{y}}{t}\right)^\top \mathbf{A} \left(\frac{\mathbf{y}}{t}\right) + 2\mathbf{b}^\top \left(\frac{\mathbf{y}}{t}\right) + c \right] \geq 0,$$

which holds true by the nonnegativity of f . □

Appendix: Classification of Matrices

Definition (Positive Definiteness). 1. A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called *positive semidefinite*, denoted by $\mathbf{A} \geq \mathbf{0}$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{R}^n$.

2. A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called *positive definite*, denoted by $\mathbf{A} > \mathbf{0}$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0} \in \mathbb{R}^n$.



Exercise: check for

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Proposition. Let $\mathbf{A}$ be a positive definite (semidefinite) matrix. Then the diagonal elements of $\mathbf{A}$ are positive (nonnegative).

Definition (Negative (Semi) Definiteness, Indefiniteness). 1. A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called negative semidefinite, denoted by $\mathbf{A} \leq \mathbf{0}$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \leq 0$ for every $\mathbf{x} \in \mathbb{R}^n$.

2. A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called negative definite, denoted by $\mathbf{A} < \mathbf{0}$, if $\mathbf{x}^\top \mathbf{A} \mathbf{x} < 0$ for every $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n$.

3. A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called indefinite if there exist $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ such that $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0, \mathbf{y}^\top \mathbf{A} \mathbf{y} < 0$.

Remark. • $\mathbf{A}$ is negative (semi)definite if and only if $-\mathbf{A}$ is positive (semi)definite.

- A matrix is indefinite if and only if it is neither positive semidefinite nor negative semidefinite.
- A symmetric matrix is with positive and negative elements in the diagonal is indefinite.
- The sum of two positive(negative) (semi)definite matrices is positive(negative) (semi)definite.

Theorem (Eigenvalue Characterization). Let $\mathbf{A}$ be a symmetric $n \times n$ matrix. Then:

- $\mathbf{A}$ is positive definite iff all its eigenvalues are positive.
- $\mathbf{A}$ is positive semidefinite iff all its eigenvalues are nonnegative.
- $\mathbf{A}$ is negative definite iff all its eigenvalues are negative.
- $\mathbf{A}$ is negative semidefinite iff all its eigenvalues are nonpositive.
- $\mathbf{A}$ is indefinite iff it has at least one positive eigenvalue and at least one negative eigenvalue.

Proof. Part a). There exists orthogonal $\mathbf{U} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{U}^\top \mathbf{A} \mathbf{U} = \mathbf{D} \equiv \text{diag} (d_1, d_2, \dots, d_n),$$

where $d_i = \lambda_i(\mathbf{A})$. Making the linear change of variables $\mathbf{x} = \mathbf{U} \mathbf{y}$, we have

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{y}^\top \mathbf{U}^\top \mathbf{A} \mathbf{U} \mathbf{y} = \mathbf{y}^\top \mathbf{D} \mathbf{y} = \sum_{i=1}^n d_i y_i^2.$$

Therefore, $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ iff

$$\sum_{i=1}^n d_i y_i^2 > 0 \text{ for any } \mathbf{y} \neq \mathbf{0}.$$

The latter holds iff $d_i > 0$ for all i . (why?) □



Trace and Determinant. Let A be a positive semidefinite (definite) matrix. Then the trace $\text{Tr}(A)$ and the determinant $\det(A)$ are nonnegative (positive). This follows directly recalling that the trace of a matrix is the sum of its eigenvalues and the determinant its product.

Exercises. Let A be a symmetric 2×2 matrix. Then A is positive semidefinite (definite) if and only if $\text{Tr}(A), \det(A) \geq 0$ ($\text{Tr}(A), \det(A) > 0$).

Classify the matrices

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0.1 \end{pmatrix}.$$

The Principal Minors Criteria

Given an $n \times n$ matrix, the determinant of the upper left $k \times k$ submatrix is called the k -th principal minor and is denoted by $D_k(A)$.

Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$D_1(A) = a_{11}, D_2(A) = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, D_3(A) = \det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Proposition. Let A be an $n \times n$ symmetric matrix. Then A is positive definite if and only if

$$D_1(A) > 0, D_2(A) > 0, \dots, D_n(A) > 0.$$

Proposition. Let A be an $n \times n$ symmetric matrix. Then A is negative definite if and only if

$$(-1)^k D_k(A) > 0, \quad \text{for all } k = 1, \dots, n.$$

This is equivalent to check that $-A$ is positive definite.

Exercise. Classify the matrices

$$A = \begin{pmatrix} 4 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 2 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & -1 \end{pmatrix}, C = \begin{pmatrix} -4 & 1 & 1 \\ 1 & -4 & 1 \\ 1 & 1 & -4 \end{pmatrix}.$$



Diagonal Dominance

Definition (Diagonal Dominance). Let A be a symmetric $n \times n$ matrix.

a) A is called *diagonally dominant* if

$$|A_{ii}| \geq \sum_{j \neq i} |A_{ij}| \quad \forall i = 1, 2, \dots, n$$

b) A is called *strictly diagonally dominant* if

$$|A_{ii}| > \sum_{j \neq i} |A_{ij}| \quad \forall i = 1, 2, \dots, n$$

Theorem (Positive definiteness of diagonally dominant matrices). a) If the matrix A is symmetric, diagonally dominant with nonnegative diagonal elements, then A is positive semidefinite.

b) If A is symmetric, strictly diagonally dominant with positive diagonal elements, then A is positive definite.

Proof. a) Suppose in contradiction that there exists a negative eigenvalue λ of A , and let u be a corresponding eigenvector. Let $i \in \{1, 2, \dots, n\}$ be an index for which $|u_i|$ is maximal among $|u_1|, |u_2|, \dots, |u_n|$. Then by the equality $Au = \lambda u$ we have

$$|A_{ii} - \lambda| \cdot |u_i| = \left| \sum_{j \neq i} A_{ij} u_j \right| \leq \left(\sum_{j \neq i} |A_{ij}| \right) |u_i| \leq |A_{ii}| |u_i|$$

implying that $|A_{ii} - \lambda| \leq |A_{ii}|$, which is a contradiction to the negativity of λ and the nonnegativity of A_{ii} .

b) since by part (a) we know that A is positive semidefinite, all we need to show is that A has no zero eigenvalues. Suppose in contradiction that there is a zero eigenvalue, meaning that there is a vector $u \neq 0$ such that $Au = 0$. Then, similarly to the proof of part (a), let $i \in \{1, 2, \dots, n\}$ be an index for which $|u_i|$ is maximal among $|u_1|, |u_2|, \dots, |u_n|$ and we obtain

$$|A_{ii}| \cdot |u_i| = \left| \sum_{j \neq i} A_{ij} u_j \right| \leq \left(\sum_{j \neq i} |A_{ij}| \right) |u_i| < |A_{ii}| |u_i|$$

which is obviously impossible, establishing the fact that A is positive definite. $\square$

